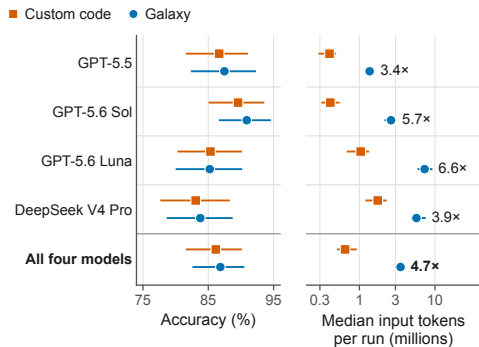


a The trade: same accuracy, more input tokens

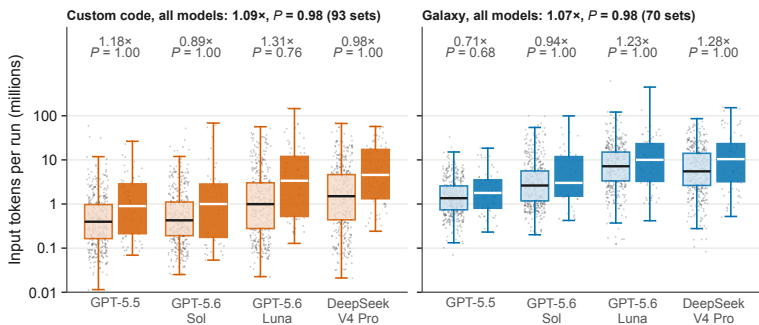
BigBench-Verified-50 and CompBioBench runs; bars, 95% confidence intervals
Numbers: times more input tokens in Galaxy on the same task



b Incorrect runs cost no more than correct runs of the same task

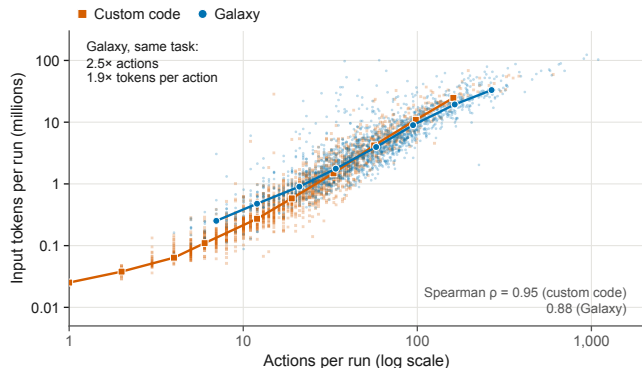
Numbers: input tokens of incorrect runs relative to correct runs of the same task and model

Custom code, correct (orange square), Custom code, incorrect (orange circle), Galaxy, correct (blue square), Galaxy, incorrect (blue circle)



c Cost follows interaction

Correct runs; lines join median input tokens in bins of actions



d Finding tools is half of what Galaxy sends back

Each Galaxy interaction is a request from the agent and a text reply from Galaxy; the model reads every reply and rereads it at each later step (all Galaxy runs)

